

Xiangru Zhou

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CAREER PROFILE

10+ years experience in image processing. I have a proven track record of innovation and excellence. My expertise spans the image processing industry, especially in programming and computer vision, where I have deep knowledge of process flows and a history of leading practical application projects to success. I've applied for over **10 patents** and am driven by the promise of technologies like Brain-Computer Interfaces (BCI) and medical image processing to improve lives.

EDUCATION

The Hong Kong Polytechnic University

MSc Microelectronics Technology & Materials

Hong Kong

Sep. 2024 – Present

Xi'an University of Technology

Bachelor of Microelectronics

Xi'an, China

Sep. 2009 – Jun. 2012

• **GPA: 3.64/5.0**

• **Average Score:** 81.18/100

• **Honours:** 2nd Prize 8th Xi'an High-tech "Challenge Cup" Shaanxi Province University Student Extracurricular Academic Science and Technology Works Competition; 1st and 3rd Prize University 19th Extracurricular Academic Science and Technology Works Competition; Special Prize & 2nd Prize University 19th "LiAo Cup"; 1st Prize University 18th "LiAo Cup"; University Innovation Achievement 2nd Prize, University 3rd Prize Scholarship

PROFESSIONAL EXPERIENCE

SMARTMORE CORPORATION LIMITED

Software Engineer

Shenzhen, China

May 2020 – Sep. 2024

AI-Driven Robotic Arm Project

Leader

May 2024 – Sep. 2024

- Branch 1: Traditional Hand-Eye Calibration + Image Recognition
 - Implemented 3D scenarios: arbitrary object recognition and grasping, sketching based on verbal commands, and interactive stacking toys with users.
 - Integrated an Intel Realsense camera, a structured light 3D camera, a 7-axis robotic arm with a gripper, two servers for large-model deployment, and a PC for control and ASR-based voice input.
 - Deployed conversational and vision models to enable seamless interaction between visual, auditory, and motor systems.
 - Achieved high precision in object localization and vision-guided robotic operations.
- Branch 2: Transformer-Based Action Prediction
 - Utilized a Transformer model to predict robotic arm actions by joint data, CNN features, and task descriptions.
 - Enabled real-time responses without hand-eye calibration, adapting to dynamic environments.
 - Tackled computational challenges and dataset requirements for effective training and deployment.

Wafer ID Reader Wafer OCR Code Reader Integrated Machine Project

Leader

Oct. 2021 – Sep. 2024

- The first deep learning-based OCR wafer character recognition tool in the semiconductor industry. Convenient use and no need to adjust the recognition parameters compare to its competing products.
- Collaborated with the algorithm team to devise strategies, conduct experiments, and select optimal algorithmic solutions.
- Modified software and SDK interfaces and functionalities based on the product manager's research and user requirements.
- Implemented end-to-end closed-loop process from requirement analysis to deployment, including evaluation, experimentation, algorithm/software/SDK development, testing, and deployment.
- Same recognition rate level as that of industry leader Cognex products in normal cases, partially exceeds its recognition rate (100% versus 99.5%) in the case of fixed format recognition.

Defect Detection System for SONY Labels

Second-hand Collaborator

Feb. 2023 – May 2023

- Conducted Proof of Concept (POC) using client samples to validate system performance.
- By combining the template image and the image to be checked together to form a two-channel image, defects are marked and trained on this basis, the accuracy has been improved by 2%.

BGI Genomics Reagent Bottle Testing

Collaborator

Oct. 2022 – Feb. 2023

- Designed algorithmic solutions and implemented the AI model into software for real-time monitoring of reagent bottle caps, ensuring seal integrity and smoothness during the manufacturing process.
- Conducted Optical Character Recognition (OCR) for character detection on reagent bottle bodies, effectively identifying characters on various colored backgrounds of medication and detecting printing defects.

Specific String OCR System for Arbitrary Orientations On Tags

Designer

Jun. 2021 – Feb. 2022

- Designed the composition of models responsible for the task, leveraging the existing MobileNet training framework in PyTorch. Trained four models individually: label localization, orientation determination, character localization, and recognition.

- Developed a software development kit (SDK) in Visual Studio to facilitate integration and usage of the OCR system.
- Designed and implemented a user-friendly interface using the QT framework to enhance the usability and accessibility of the OCR system.

Character Recognition System for Apple Watch Bands Using Laser Engraving Technology

Designer

May. 2020 – May 2021

- Trained character localization and classification models on a four-card 2080Ti server running Ubuntu, utilizing the MobileNet architecture.
- Employed the ONNX Runtime (ORT) for model deployment, integrating a C++ SDK for inference on the server and developing a user-friendly software interface.
- Collaborated with colleagues to align and optimize the SDK, including pre-processing and post-processing workflows.
- Precision and recall were above 99.7% respectively.

Defect Detection System for Apple Watch Bezels Using 3D Imaging Technology

Designer

Jun. 2020 – Jun. 2020

- Implemented the conversion of 3D point cloud data into 2D heatmaps for analysis purposes.
- Designed and programmed the user interface for the software application.

SHENZHEN PHDI CORPORATION LIMITED

Co-founder, CTO

Shenzhen, China

Jan. 2017 – May 2020

Android Software for A Car-Mounted Mobile Phone Controller

Designer

Jan. 2017 – May 2020

- Managed PCB production designed Bluetooth chip-related programs and conducted 3D design and printing for the product's appearance.
- Enabled rapid switching between navigation, music, and WeChat applications in a car environment.
- Implemented automatic sending and receiving of WeChat messages.
- Integrated features for one-touch activation of voice navigation and song search.
- Designed and developed a comprehensive product including a mobile application and a Bluetooth controller.

Face Recognition and Tracking Software on The Android Platform

Designer

Jan. 2017 – May 2017

- Utilized OpenCV, Caffe, Idlib, and VGG net deep learning neural network for face localization and gender recognition.
- Integrated BP neural network for gender recognition within the face detection algorithm.
- Applied the developed algorithm in systems for estimating the attention of different gender groups towards billboards and for access control systems, resulting in the acquisition of a patent.
- Adapted the algorithm for integration with security cameras to track familiar/strange faces indoors, aiding in the detection of potential security threats.
- Successfully delivered the developed solution to a renowned domestic security camera company.

OPT MACHINE VISION TECH CO., LTD.

Image processing algorithm engineer

Guangdong, China

Apr. 2014 – Jan. 2017

One-Dimensional Barcode Localization and Recognition Algorithm

Designer

Apr. 2014 – Jan. 2017

- Achieved automatic localization of barcodes in approximately 10ms on a PC equipped with an i3 processor, capable of identifying barcodes in complex images of up to 2 million pixels.

Two-Dimensional Barcode Localization and Recognition Algorithm

Designer

Jan. 2015 – Jan. 2017

- Implemented functionality to locate and recognize QRCode, DataMatrix, and other two-dimensional barcodes.
- Utilized Google's open-source libraries ZXing/Zbar and made necessary modifications for enhanced performance

Rapid Edge And Circle Detection Algorithm

Designer

Jan. 2015 – Jan. 2017

- Utilized non-Hough fitting method to swiftly fit specified parameters of lines and circles in designated directions, implementing the algorithm to effectively recognize lines and circles in blurry edge images of up to 2 million pixels.
- Achieved real-time identification within 10ms on a PC with an i3 processor and 8GB of memory.

ADDITIONAL INFORMATION

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- **Language:** Cantonese, Mandarin, English (Fluent; IELTS: 6.5), Japanese (N5)
 - **Programming Language:** C, C++, Python, Java, Java Script, HTML, CSS, PHP
 - **Development Tools & Libraries:** QT, Visual Studio, gdb/pdb, Git, Docker, Anaconda & OpenCV, PyTorch, Halcon
 - **Other Skills:** Keil (STM32), Altium Designer (PCB), Rhino (3D printing), Dreamweaver (Website)
 - **Books:** Computer Vision: Algorithms and Applications, Design Patterns: Elements of Reusable Object-Oriented Software
 - **Patents:** CN202311136494.1; CN202310973415.6; CN202310287581.0; CN202310051665.4; CN202211602503.7; CN202211474648.3; CN202211357774.0; CN202210994137.8; CN202230264395.1; CN201911330078.9; CN201110120104; CN201110308514.X